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AGE STRUCTURE AND SOME GROWTH PARAMETERS IN A POPULATION OF *Bombina bombina* (L., 1761) FROM TURKEY

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We studied age, growth and longevity in a population (Değirmenyeni) of *B. bombina* from Turkey by skeletochronology performed on the phalanges. The mean age was found to be 4.23 years. The maximum life span was 7 years. The sexual maturity was 2 – 4 years. The sexes did not exhibit any differences in terms of mean body size (SVL). For both sexes, we found a positive correlation between SVL and age. Growth pattern estimated by von Bertalanffy's showed a good fit to the relation between age and SVL. For both sexes, the estimated asymptotic SVL was smaller than the maximum SVL recorded. In conclusion, our results show similar growth rates between both sexes having similar longevity and age at sexual maturity.

Keywords: fire-bellied toad; longevity; skeletochronology; thrace; Turkey.

INTRODUCTION

The fire-bellied toad, *Bombina bombina* (L., 1761), is a member of genus *Bombina* having eight species (<http://amphibiaweb.org>). It is distributed in Eastern and Central Europe, the Balkans, and west Anatolia (<http://amphibiaweb.org>). Its Turkish distribution is limited to Marmara region. *B. bombina* has been classified in the LC (Least Concern) category in the IUCN Red List. Although *B. bombina* is one of the less common animals within the anuran species in Turkey, there is no detailed study based on population dynamics (age and growth) of the species. Age determination has been used to obtain some information about population dynamics and life histories of amphibian species by researchers for many years (Castanet and Baez, 1991; Caetano and Castanet, 1993; Smirina, 1994; Miaud et al., 2001; Cogălniceanu and Miaud, 2003; Esteban et al., 2004; Guarino et al., 2008; Patrelle et al., 2012).

Skeletochronology is known to be an effective and reliable method to estimate the age of amphibians (Castanet and Smirina, 1990). There are some age-determination studies on Turkish amphibians (Kutrup et al., 2005, 2011a, 2011b; Olgun et al., 2005; Yılmaz et al., 2005;

Üzüm, 2009; Üzüm and Olgun, 2009; Gül et al., 2011; Özdemir et al., 2012). However, there is no comprehensive study based on age and growth of *B. bombina* in Turkey while age structure and body size of the species were studied in Russia, Denmark and Romania (Shaldybin, 1976; Ananjeva and Borkin, 1979; Briggs, 1996; Cogălniceanu and Miaud, 2003), respectively.

The aim of the present study was to observe the age structure and some growth parameters in a population of *B. bombina* from Turkey by skeletochronology.

MATERIAL AND METHODS

A total of 35 specimens (20 ♂♂, 15 ♀♀) were caught from a population in Değirmenyeni (Edirne) (number of permission to capture is: 72784983-488.04-147332). The Değirmenyeni population (41°45'39" N 26°33'43" E) is located in a lowland area at the 38 m a.s.l. altitude. In habitat of the population, there are small puddles which are comprised from small water streams used by farmers. The average water temperatures were measured as 21°C during the sampling period (from 1 to 5 September 2015). In general, the active period for the toads changes from early April to October in Değirmenyeni population.

Within each site, the toads were caught by hand and sexed by direct examination of the secondary sex charac-

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teristics (nuptial pads of the males). Snout-vent length (SVL) was measured with a digital caliper. We quantified Sexual Size Dimorphism (SSD) with the Lovich and Gibbons (1992) index according to the following formula:

$$SDI = \frac{\text{mean length of the larger sex}}{\text{mean length of the smaller sex}} - 1$$

(resulted number must be less 1). When arbitrarily add minus if males are larger females.

For each toad, the second phalange of the longest finger from the hind limb was clipped and preserved in 10% formaldehyde solution for subsequent histological analyses. After registration and toe-clipping, the toads were released back into their natural habitats (permission number of animal care and ethics committee: KTÜ.53488718-264/2015/17).

The procedure of skeletochronology is based on a calculation of the lines of arrested growth (LAGs) in the toe's bone tissue of the toads. After skins of the toes were peeled, they were waited 2.5 h in 5% nitric acid solution for decalcification of bone tissue. Later, toe samples were loaded into the tissue processing system (Tissue processor with Leica brand). Then, all tissue samples were embedded in paraffin with a tissue embedding device (Thermo brand). The cross-sections (15 µm) were obtained from embedded phalanges with rotary microtome, after which they were stained using the hematoxylin procedure (Bülbül et al., 2016). The stained cross-sections, which were put on the microscope slides, were closed using entellan. Cross-sections covered by entellan were observed under a light microscope.

The numbers of LAGs on the cross-sections were independently calculated by three observers (U. Bülbül, A. İ. Eroğlu, and M. Kurnaz). The observed double lines were not taken into account for age determination. As previously stated in the study of Bülbül et al. (2016), we assessed endosteal resorption of the first LAG by comparing the diameters of eroded marrow cavities with the diameters of noneroded marrow cavities in sections from the youngest specimens. The distance between two adjoining LAGs is a good indicator of individual growth in a given year (Kleinenberg and Smirina, 1969; Özdemir et al., 2012). Where we observed an obvious decrease in spacing between two subsequent LAGs, we took it to mark the age when sexual maturity was achieved (Ryser, 1998; Yılmaz et al., 2005; Özdemir et al., 2012). Any serious problems depend on age estimation was not caused by the endosteal resorption.

Because age classes and body measurements (SVL) were normally distributed (one-sample Kolmogorov – Smirnov's test, $P > 0.05$) we used parametric tests for comparison of means (independent sample t -test) and for correlation (Pearson's correlations test, $P < 0.05$). All

statistic tests were processed with SPSS 21.0 for Windows and the level of significance chosen was $P < 0.05$.

Using the von Bertalanffy's model, we formed the growth curves as described in previous literature for anurans (Esteban et al. 2004; Cogălniceanu and Miaud, 2003; Guarino et al., 2011; Kutrup et al., 2011a). The general form of the von Bertalanffy growth equation used is

$$L_t = L_\infty [1 - e^{-k(t-t_0)}],$$

where L_t is length at age t , L_∞ is a parameter depicting asymptotic maximum length, k is a growth coefficient, and t_0 is the age at hatching, which is the starting point of the growth interval under the present study. Because data on the size at hatching of the studied population were lacking, we assumed as size at hatching ($L_{t_0} = 22.6$ mm) the mean value provided by Cogălniceanu and Miaud (2004) because of the lack of incontrovertible data on the size at hatching of the studied populations, due to lack of young toads collected during the field studies. The parameters L_∞ (asymptotic SVL) and k , and their asymptotic confidence intervals (CI), were estimated using a non-linear regression procedure by means of the IBM SPSS 21.0 software program. Then, the growth rates were calculated as $R = k(L_\infty - L_t)$. Growth curves were considered to be significantly different if the 95% confidence intervals did not overlap (James, 1991; Wapstra et al., 2001).

RESULTS

The means of SVL and age values were 34.7 ± 0.6 mm and 4.2 ± 0.2 years for the Değirmenyeni population (34.6 ± 0.9 mm; 4.10 ± 0.28 years in male specimens and 34.9 ± 0.7 mm; 4.47 ± 0.27 years in female specimens), respectively. Descriptive statistics of Değirmenyeni population are given in Table 1.

A growth zone and thin hematoxylinophilic outer line corresponding to a winter line of arrested growth were present in cross sections of the phalanges in 92.5% ($n = 37$) of *B. bombina* (Fig. 1). The resorption zone reached the first LAG in 5 specimens (13.51%) for *B. bombina* but never created difficulty for age determination. We observed double lines in 6 (17.14%) specimens of *B. bombina*. The oldest females and males in Değirmenyeni population were 7 and 6 years old, respectively (Fig. 2). The age at maturity was 2 years in 26 (74.29%) specimens, 3 years in 8 (22.86%) specimens and 4 years in 1 (2.85%) specimens of the *B. bombina*.

Age was ranged from 3 – 7 years in females and 2 – 6 years for males. Age of specimens from Değirmenyeni population did not differ significantly between the sexes (independent sample t -test; $t = -0.733$, $df = 33$,

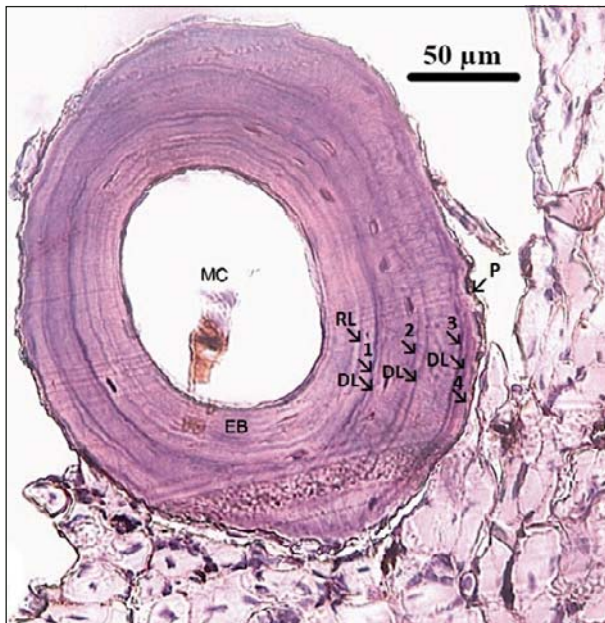


Fig. 1. Cross-sections (15 μm thick) through phalanges of four-year-old female (35.6 mm SVL) *Bombina bombina*. For abbreviations see the text; MC, marrow cavity; EB, endosteal bone; RL, resorption line; P, periphery; DL, double line.

$P = 0.469$). Intersexual difference in body size (length) was female-biased ($\text{SDI} = 0.009$). However, both SVL ($t = -0.300$, $df = 33$, $P = 0.766$) and body mass ($t = 0.166$, $df = 33$, $P = 0.869$) did not differ significantly between sexes. There was no correlation between SVL and age for males (Pearson's correlation $r = 0.302$, $P = 0.195$) and females ($r = 0.384$, $P = 0.158$).

Growth pattern estimated by von Bertalanffy's showed a good fit to the relation between age and SVL (Fig. 3). For both sexes, the estimated asymptotic SVL was smaller than the maximum SVL recorded ($\text{SVL}_{\text{asym}} \pm \text{CI}$, males: 35 ± 8 mm; females: 35 ± 9 mm). Growth coefficient was lower in males than in females ($k \pm \text{CI}$, males: 1.5 ± 0.2 ; females: 1.7 ± 0.3). The growth curve of males was slightly different from that of females. Growth rates of the males and females were not significantly different within the population (Independent Sample t -test; $t = 1.195$, $df = 8$, $P = 0.266$).

DISCUSSION

We provide here the demographic data on a *B. bombina* population present in Turkey while there are some studies related to the age structure of *B. bombina* population from different countries.

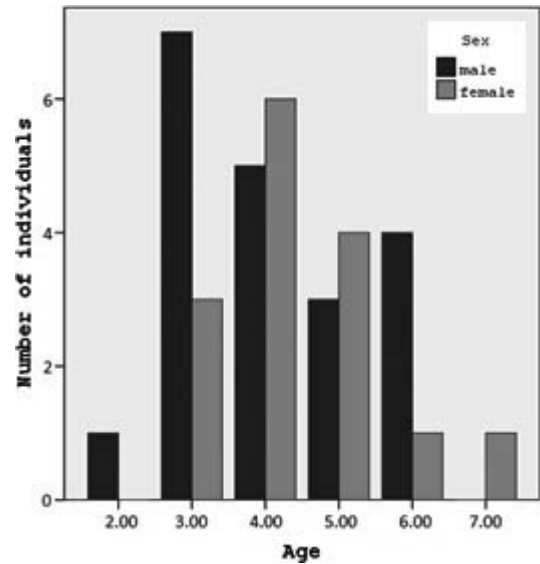


Fig. 2. Age distribution of adult specimens of *Bombina bombina*.

Maximum longevity was found as 12 years in Denmark (Briggs, 1996), 11 years in Russia (Ananjeva and Borkin, 1979; Smirina, 1994) and 5 years in Romania (Cogălniceanu and Miaud, 2003) for *B. b. bombina* while we found maximum life span 7 years in the Değirmeniyeni population of *B. bombina* from Turkey. The longevity changes according to the altitude, latitude and other climatic and environmental factors (Berven, 1982; Miaud et al., 1999). Because the annual average temperature in Denmark and Russia is lower than Turkey and Romania, the toads living in colder climates (in Denmark and Russia) tended to reach higher longevity. The active period of the toads in cold regions is short. Thus, the toads mature later in colder climates and their longevity is higher than those living in the hot climate. A different climatic condition might affect population demography by generating differences in age structure and longevity (Özdemir et al., 2012). On the other hand, Cogălniceanu

TABLE 1. Descriptive Statistics of Age, SVL, and Body Weight of the Değirmeniyeni (*B. bombina*) Population.

Charac- ters	Sex	<i>n</i>	Mean	Range	SE	SD
SVL	♂♂	20	34.6	28.3 – 44.3	0.9	3.83
Age		20	4.1	2 – 6	0.3	1.25
SVL	♀♀	15	34.9	30.1 – 38.0	0.7	2.64
Age		15	4.5	3 – 7	0.3	1.06
SVL	♂♂	35	34.7	28.3 – 44.3	0.6	3.34
Age	♀♀	35	4.2	2 – 7	0.2	1.19

Note. For abbreviations, see the text; *n*, number of samples; SD, standard deviation; SE, standard error.

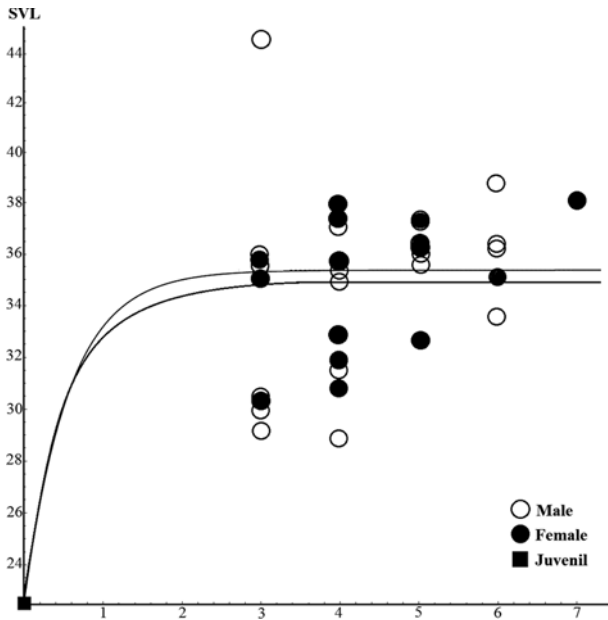


Fig. 3. The von Bertalanffy growth curves for *Bombina bombina* (open circle, solid line for males and solid circle, blue line for females). The solid square shows SVL mean (22.6 mm) of the juvenile toads at hatching as reported by Cogălniceanu and Miaud (2004). Growth parameters are given in the text.

and Miaud (2003) explained the lower observed longevity for *B. bombina* because of the floodplain environment in Romania. However, we cannot consider a similar environment for Değirmenyeni population.

The SVLs of individuals in Değirmenyeni population was similar to the length range of *B. bombina* reported by Cogălniceanu and Miaud (2003) in Romania. SVL of adult males varied from 28.3 – 44.4 to 30.1 – 38.0 mm in females in the Değirmenyeni population, while these values were determined as 30.3 – 40.0 mm in males and 30.5 – 41.0 in females for Romania population. However, Fog (1996) and Rafińska (1991) reported higher sized individuals of *B. bombina* species in Denmark and Poland, respectively (the maximum SVL was recorded as 50 mm from populations in Denmark and females of the small toad *B. bombina* in Poland had a mean SVL of 47.1 mm). Different body size values among the individuals of the species may be related to differences in temperature. In anurans, body size is expected to increase in colder areas, being inversely related to the climatic factors (i.e., heat and sunlight) directly conditioning individual thermoregulation (Hutchinson and Dupre, 1992).

In our study, a low level of male-biased sexual size dimorphism (SSD) was observed in Değirmenyeni population. But these differences were not statistically significant. The non-significant differences in SSD may be due

to the similarity in longevity and age at maturation in both sexes in the population. Longevity and age at first reproduction have been identified as the main determinants of SSD at an intra-specific level (Liao and Lu, 2010; Lyapkov et al., 2010). In addition to, a delayed sexual maturity would allow females to reach a larger asymptotic size than males (i.e., female-biased SSD; Hasumi, 2010). Similar to this explanation, the asymptotic size of the females of Değirmenyeni population was found larger than the males of the same population. Another effective factor on SSD is sexual differences in the growth rates. In general, larger sex grows faster than smaller sex (John-Adler and Cox, 2007; Kolarov et al., 2010). Congruently we found no significant difference between growth rates in both sexes in Değirmenyeni population and the observed male-biased SSD was very low.

In the present study, the age of adult toads was not correlated with their body size (SVL). Contrary to our findings, Cogălniceanu and Miaud (2003) found a significant correlation between SVL and age of a *B. bombina* population in Romania. In amphibians, the adult body size depends on many factors, including tadpole growth rate, time of and size at metamorphosis, juvenile and adult growth rates, age at maturity and longevity (Özdemir et al., 2012). According to the literature (Monnet and Chery, 2002; Matthews and Miaud, 2007; Kutrup et al., 2011a), male frogs often mature earlier than females. However; age at maturity was found similar for both sexes (2 years) in our study. This may be due to the similarity in growth of both sexes. Congruently, SVL did not differ significantly between sexes within Değirmenyeni population. Consistent with our findings related to sexual maturity, Cogălniceanu and Miaud (2003) reported that age at maturity was found similar for both sexes (2 years in both males and females) within a *B. bombina* population in Romania. In amphibians, maturity is generally attained after they have reached a certain minimum size (Gibbons and McCarthy, 1983; Hemelaar, 1988; Ryser, 1988). On the other hand, Olgun et al. (2005) stated that age at maturity and longevity were greatly influenced by local conditions (i.e., reduction of the length of the active period).

We observed double lines in 6 (17.14%) specimens of *B. bombina*. Double lines are irregularities in bone deposition caused by unpredictable ecological factors, such as dry period, hot climate, food availability and other conditions (Jakob et al., 2002; Guarino and Erişmiş, 2008; Özdemir et al., 2012). Because the toads used in our study have not inhabited in an arid area, we did not find any negative effects of the arid period on growth, leading a higher percentage of double lines.

In addition to double lines, endosteal resorption in bone specimens also may be related to environmental

conditions (Smirina, 1972). For example, Caetano and Castanet (1993) reported that fewer resorptions were observed in lowland populations than in populations living at high altitudes. Similar to their explanation, we found the lower percentage of endosteal resorption in the Değirmenyeni population of *B. bombina* living at a lower altitude (13.51% for *B. bombina*). However, a converse trend was reported in two water frog species; *Rana perezi* and *Rana saharica* (Esteban et al., 1996, 1999), respectively. On the other hand, daily and annual activity and climate conditions probably have a strong effect on the intensity of bone resorption of anuran long bones (Hemelaar, 1988; Esteban, 1990; Leclair, 1990; Augert, 1992; Esteban et al., 1999).

In conclusion, our results show similar growth rates between both sexes having similar longevity and age at sexual maturity. The present study provides important data about age structure and some growth parameters of *B. bombina*. Age composition of a population in amphibians varies depending on the environment. In order to conduct a powerful statistical analysis to test which factor (temperature, age at maturity, longevity and growth rates) is more effective on SSD of *B. bombina*, further long-term and controlled studies are necessary, as well as comparisons with toads from different environments.

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